

Day 10 Engineering Summary Sheet

Advanced Terraform Architecture: Loops, Modules, and Cryptographic Authentication

This technical summary details the modular architectural progression completed during the Day 10 engineering lab, focusing on scalable multi-environment provisioning loops, overcoming public cloud boundary constraints, and implementing secure cryptographic handshake access models.

1. Architectural Evolution: Monolith to Loops

The core objective of today's lab was migrating away from standard duplicate declarations to an optimized loop pattern. Using the `for_each` expression, a single custom-designed AWS architecture infrastructure block was instantiated to deploy completely isolated dual-stack environments simultaneously.

Environment Scope	Logical Key	VPC Network CIDR	Public Subnet CIDR
Testing Target	testing-1	10.11.0.0/16	10.11.1.0/24
Production Target	production-1	10.21.0.0/16	10.21.1.0/24

2. Resolving Cloud Limits & Operational Constraints

- **VpcLimitExceeded Error:** Discovered a core cloud constraint where AWS restricts standard accounts to a regional quota ceiling of $V = 5$ active Virtual Private Clouds (VPCs).
- **State Resolution:** Tearing down previous resource paths via `terraform destroy` freed allocated resources and cleared out orphaned network layers.
- **Dynamic Scaled Labs:** Downscaling the environment dictionary mapping matrix allowed successful execution without hitting cloud limit constraints while maintaining the loop framework.

3. The Cryptographic Handshake Model

Securing cloud compute layers requires migrating fully away from standard password configurations to asymmetric key cryptography workflows. The handshake operates under a dual-token paradigm:

🔑 The Cryptographic Relationship Model

The connection validation follows the mathematical equivalence function:

$$f(\text{Key}_{\{\text{Private}\}}, \text{Lock}_{\{\text{Public}\}}) \rightarrow \text{Access}_{\{\text{Granted}\}}$$

- **Public Key (The Lock):** Stamped inside the target instance's `~/.ssh/authorized_keys` security subsystem during the initial runtime lifecycle deployment by Terraform.
- **Private Key (The Physical Key):** Stored securely locally on the management client terminal under a highly protected path (e.g., `~/.ssh/finland-key.pem`).

4. Core Execution Blueprint Command Log

Below is the structured operational command line record used to execute the architecture:

```
# Verify configuration files for code grammar compliance
terraform validate

# Plan the multi-environment structural infrastructure expansion
terraform plan

# Deploy the scaled environments dynamically across the execution matrix
terraform apply

# Securely query and connect to live instances utilizing target routing paths
ssh -i ~/.ssh/finland-key.pem ubuntu@56.228.9.158
```

End of Day 10 Systems Engineering Log. Built and validated successfully.